

# Solution Architecture

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A Comprehensive Measure of Well-Being | HDI Predictor

## Frontend Architecture

- TanStack Router — file-based routing (/routes/index.tsx)
- React State (useState, useMemo) — form & result management
- computeHDI() — UNDP formula implementation
- tierOf() — score to tier classification
- Tailwind CSS — responsive styling

## Component Hierarchy

- → →
- ■■■ Country with 20 presets
- ■■■ components (Life, MYS, EYS, GNI)
- ■■■ Predict → computeHDI()
- ■■■ Result display with tier color

## Deployment Architecture

- GitHub repo → Vercel CI/CD → Edge Network
- Auto-deploy on every push to main branch
- Global CDN for fast worldwide access

## Flask Architecture (ML version)

- HDI.csv → Preprocessing → LinearRegression → model.pkl
- Flask app.py → /predict route → render resultnew.html